

PROJECT REPORT

Wheat Price Forecasting using Machine Learning

1. Introduction

Machine Learning is widely used for prediction tasks in various fields such as finance, healthcare, and agriculture. In agriculture, price prediction plays a crucial role in helping farmers and traders make better decisions.

Crop prices fluctuate due to multiple factors such as demand, supply, and seasonal variations.

This project focuses on predicting wheat prices using historical mandi data with the help of machine learning techniques.

This project demonstrates the practical application of data science in solving real-world agricultural problems.

2. Problem Statement

Wheat prices fluctuate daily due to supply and demand conditions in the market. Farmers often sell their produce without knowing future prices, which may lead to financial losses.

There is a need for a system that can analyze historical data and predict future wheat prices.

A predictive model can help reduce uncertainty and improve decision-making in agricultural markets.

3. Dataset Description

The dataset used in this project is taken from government sources such as AGMARKNET and data.gov.in.

It contains real mandi-level data for wheat prices in Haryana.

The dataset includes the following features:

- Arrival Date

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- Modal Price
- Arrival Quantity

The dataset reflects real market behavior and provides a reliable base for building predictive models.

4. Data Preprocessing

The following preprocessing steps were performed:

- Removed null and missing values
- Converted price values from string to numeric format
- Converted date column into datetime format
- Created a new feature called “Day” for model training
- Cleaned and formatted arrival quantity values

Proper preprocessing ensures that the model learns meaningful patterns from clean and structured data.

5. Methodology

Different machine learning models were applied to predict wheat prices:

- Linear Regression
- Multiple Linear Regression
- Random Forest Regressor

Random Forest was selected as the final model because it provides better accuracy and can handle non-linear relationships in data.

Random Forest is an ensemble learning technique that improves prediction performance by combining multiple decision trees.

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6. Model Evaluation

The performance of the models was evaluated using Mean Absolute Error (MAE).

MAE measures the average difference between actual and predicted values.

Results:

- Linear Regression: Higher error
- Random Forest: Lower error

Lower MAE indicates that the predictions are closer to the actual values, making the model more reliable.

7. Results

The model successfully predicts wheat prices based on historical data.

It provides the following capabilities:

- Prediction for any future date
- Multi-day and yearly forecasts
- Visualization using graphs

It was observed that price trends are influenced by arrival quantity, indicating a supply-demand relationship.

8. Conclusion

This project demonstrates that machine learning can be effectively used for crop price prediction.

The Random Forest model provided better results compared to Linear Regression.

The system helps in understanding price trends and supports better decision-making.

The project converts historical data into actionable insights for real-world applications.

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9. Limitations

- The dataset is limited to a few months
- External factors such as weather and demand are not included
- Prediction accuracy can be improved with more data

Increasing dataset size and adding more features can significantly improve model performance.

10. Future Scope

The project can be further improved by:

- Adding more historical data
- Including weather and demand-related features
- Using advanced models such as LSTM
- Developing a real-time web or mobile application

This system can be extended into a real-time agricultural decision support tool.

NAME CHIRAG GARG

SAPID 500121978